

Name:

Date:

Period:

Car Lab #1

Purpose

To determine the ways in which a position graph represents an object's motion.

Materials

- Pick up one blue car and one red car
- Pick up a dry erase marker and an eraser
- Pick up meter stick
- Pick up a timer

Procedure

Before you get started, measure the time it takes each car to travel 1.00 meter. Record the time and calculate the velocity of each car using the equation $v = \Delta x / \Delta t$.

Red Car	
Time to travel 1.00 m (s)	Velocity (m/s)

Blue Car	
Time to travel 1.00 m (s)	Velocity (m/s)

When you are ready to start the experiment:

1. Make a "track" that is 5 meters long. Draw a line and label at "0 m," "1 m," "2 m," "3 m," "4 m," and "5 m."
2. Put the car at 0 m. Count down, release the car, and have the timer start calling out the time every 2 seconds.
3. Every 2 seconds for 12 seconds, make a mark at the location of the car.
4. Once the car has finished its trip, measure each 2-second mark relative to the zero point.
5. Repeat 3 times
6. Repeat for the other car

Data

Experiment #1: Red Car				
Time (s)	Displacement (m)			
	Trial #1	Trial #2	Trial #3	Average
2				
4				
6				
8				
10				
12				

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Experiment #2: Blue Car				
Time (s)	Displacement (m)			
	Trial #1	Trial #2	Trial #3	Average
2				
4				
6				
8				
10				
12				

Analysis

Make a graph on Desmos of your two data sets **Time** be x , **Average displacement** be y . Plot a best fit line. Write down the equation for each of the best fit lines below. You should also copy the link to your graph, paste it on Schoology, and include the link here.

Equation for Red Car:

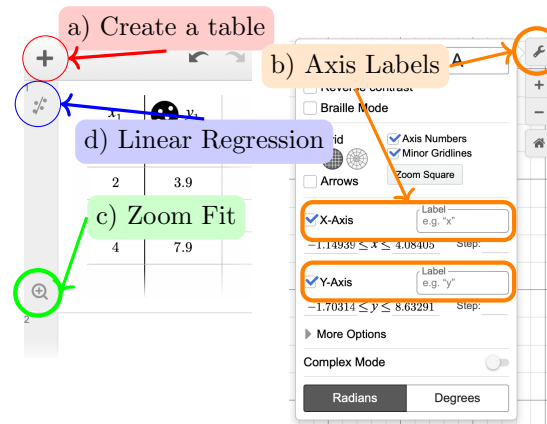
Equation for Blue Car:

Link to graph: <https://www.desmos.com/calculator/>_____

How to Graph on Desmos

Go to www.desmos.com/calculator to graph your data.

- Start by making a table by clicking the “+” icon at the top left. You will need to create two separate tables.
- Make sure to label the axes using the wrench icon at the right.
- Zoom out so that you can see the whole graph and so that it fills the page. You can do this using the “Zoom Fit” option, but be careful that your fit does not cut off one of the graphs
- Create best fit lines for each graph using the “Linear Regression” tool
- Copy a link to your graph using the export button and clicking “Share a Snapshot”. Write down the link to this graph so you and your instructor can both access your work, later.



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Write a paragraph explaining what you did on this part of the lab. Your paragraph should address: (1) what you graphed and what the graph looked like, (2) why the shape of the graph makes sense, and (3) what the physical meaning is of the *slope* and the *y-intercept*.

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Conclusion

You calculated the velocity of each car on the first page. Compare the velocities with the slopes you calculated for each car by calculating a percent error.

$$\text{Percent Error} = \frac{|\text{measured} - \text{expected}|}{\text{expected}} \times 100\%$$

Calculate the percent error and then write a paragraph commenting on your percent error. Your paragraph should address: (1) How reasonable is your percent error? (2) Identify three to four factors that could have created uncertainties in your measurements and explain how each factor may have influenced your percent error. (3) What can you say you learned about the position graphs of objects and how they are related to the motion of those objects?

Percent Error (Red Car): _____ Percent Error (Blue Car): _____